

A study of two-phase annular flow using unsteady numerical computations

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ARTICLE INFO

Article history:

Received 30 October 2018

Revised 7 April 2019

Accepted 7 May 2019

Available online 8 May 2019

Keywords:

Annular flow

Passive scalar mixing

Liquid films

CFD

Subchannel

ABSTRACT

Flow behavior of thin liquid films in an annular flow regime is an important element of the thermal hydraulics of a BWR – it controls the heat transfer from the fuel rods to the coolant. Despite its relevance, the flow behavior of dynamic liquid films is not well understood even in adiabatic conditions. To investigate it, unsteady numerical simulations with an interface tracking model were performed in a double subchannel geometry with a P/D ratio of 1.325. Different turbulence models, namely the large eddy simulation (LES) and linear/nonlinear eddy viscosity unsteady-RANS (URANS) models, were tested. A novel approach for generating turbulent inlet conditions in a periodic flow domain was developed to reduce computational efforts. Validation against experimental data revealed shortcomings of the linear eddy-viscosity RANS model in predicting key flow parameters. By capturing the effects of the secondary flow structures in a subchannel geometry, improved predictions were obtained with a non-linear SST $k-\omega$ (QCR) turbulence model. Time-averaged liquid film thickness (LFT) and tracer distribution obtained with LES were found to have the best agreement with experimental data. Instantaneous and time-averaged velocity profiles were analyzed to understand the influence of the gas-liquid interface. Secondary flow structures in the subchannel gap region were found to enhance the turbulent mixing of the passive scalar in the liquid film. This finding is relevant towards the prediction of thermal-hydraulic parameters in a multichannel flow assembly by accounting for inter-channel mixing phenomena.

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1. Introduction

Two-phase annular flow regime is established in boiling water reactor (BWR) subchannels where a high steam quality is present due to coolant boiling. The avoidance of critical heat flux (CHF) is paramount in the normal functioning of the reactor, which can lead to dryout of liquid film on the fuel rod surface. Under such flow conditions, the temperature of the fuel rod may increase substantially due to the reduced heat transfer coefficient in a steam-only environment. This can lead to fuel-cladding oxidation and an eventual failure of its mechanical integrity. It is important, from an engineering standpoint, to acquire a deeper understanding of the flow to achieve better fuel utilization in terms of thermal margins while preventing the occurrence of dryout on the fuel rods.

The annular flow regime consists of a thin liquid film on the wall characterized by a dynamic interface and a gas core in the center of the flow channel. Each phase may also be transported in

a dispersed form either as liquid droplets in the gas core or gas bubbles in the liquid film (Fig. 1). The transfer of heat through the liquid film is modulated by the flow properties of the gas-liquid interface through the action of the so-called ripple and disturbance waves, respectively. The waves are understood to play a significant role in the heat and mass transfer in annular flow regime, for example as shown in the work of Jayanti and Hewitt (1997). Hence, it is important to model this characteristic unsteady behavior of the flow.

Large velocities and high Reynolds numbers characterize the annular flow regime, which makes it highly challenging to simulate. Hence, only a few numerical studies have been undertaken in this field. A lack of highly resolved experimental data for model validation is also another reason. Fukano and Inatomi (2003) performed DNS simulations of horizontal annular flow in a pipe to study the influence of disturbance waves on the liquid film spreading in the circumferential direction. In their DNS study, the grid resolution employed was much coarser than that used in the current study for similar superficial velocities of the two-phases and, therefore, not all physical phenomena might have been accurately captured. Rodriguez (2009) performed a DNS study of annular flow

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Nomenclature

k	turbulent kinetic energy
D_h	hydraulic diameter
u_τ	shear velocity
μ_t	turbulent viscosity
u_n	normal velocity
τ_w	wall shear
ρ	density
y	distance from wall
Φ_i	phase volume fraction
J_ϕ	phase superficial velocity
$\dot{m}_\phi$	phasic mass flow rate
y^+	non-dimensional wall distance
u^+	non-dimensional streamwise velocity

to aid in the development of physics-based closure models used in a multi-field modeling approach. The work was able to capture the major mechanisms observed in annular flow but was restricted to an angular section of a pipe.

Anglart and Caraghiaur (2011) reviewed an Euler–Euler and Euler–Lagrange approach to model boiling annular-mist flow while Liu et al. (2011) simulated roll waves in annular flow. Saxena et al. (2016) simulated annular flow in a tight lattice sub-channel geometry using VOF and got a good agreement with neutron tomography data. Recently, Sato and Niceno (2017) performed an LES of upward annular boiling flow on a flat surface with an interface tracking method to study the properties of disturbance waves. The work of Carlucci et al. (2004) summarizes the correlations and generalized relationships for single and two-phase turbulent mixing across both vertical and horizontal interconnected flow channels. Highly resolved CFD work on passive scalar mixing in two-phase annular flow is still lacking to date.

The primary objective of this work is to test the performance of numerical models available in commercial two-phase CFD codes by validating the liquid film distribution on a half-rod placed in a double subchannel geometry and tracer mixing behavior within this film. The presence of a gap between the two flow channels allows for the study of flow phenomenon responsible for mass and momentum exchange between them. The simulations were performed with the code STAR-CCM+ v. 11.06. An experimental database of liquid film flow from Damsohn and Prasser (2010), and passive scalar mixing from Saxena and Prasser (2016), is used for numerical validation. The numerical estimation of the spreading rate of the passive scalar in liquid film permits a direct validation of the model by comparing against experimental tracer data. The use of unsteady turbulence models, in the form of LES and URANS, has been made to obtain time-resolved flow features, particularly with regards to investigating the dynamic gas-liquid interface. The high spatio-temporal resolution of the numerical data also allowed for a more detailed analysis of the liquid film flow properties, which was not possible with the experimental setup. While a single experimental test-case is simulated due to the high computational resources required, a comparison of different flow parameters, both time-averaged and time-resolved, is done to discuss the current standing of common modeling approaches for simulating complex multiphase flows.

Prior to the description of the modeling strategy, a few details of the experimental setup are provided to help establish the background behind this work. Experimental data was made available from two-phase annular flow experiments performed at ETH Zurich in an adiabatic setting in a vertical, double-subchannel geometry. The channel length of 2.5 m was equivalent to a flow development length of 70 hydraulic diameters. The cross-section of

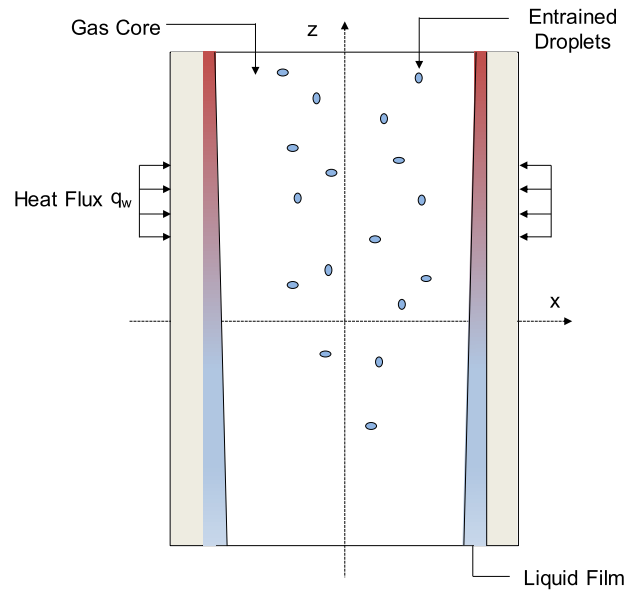


Fig. 1. Schematic representation of two-phase annular flow regime.

Table 1
Main flow parameters.

D (mm)	P (mm)	D_h (mm)	J_g (m/s)	J_l (m/s)
20	26.5	18.85	30	0.1

the flow geometry is shown in Fig. 2. The liquid was injected through a special slit, 0.5 mm in thickness, directly on the mock fuel-rods. This facilitated in orienting the flow around the rods resulting in a faster flow development. It also helped define the boundary conditions conveniently on the inlet of the simulation domain. Flow properties of the liquid film were studied by injecting a highly-conductive passive scalar. It was tracked using a liquid film sensor (LFS) which is able to detect changes in the conductance of the medium in contact with it (Damsohn and Prasser, 2009). With a measurement grid of 64×16 points around a half-fuel rod, the sensor is able to provide a two-dimensional mapping of the conductance distribution with a frequency as high as 10 KHz. Both time-resolved and time-averaged flow properties were obtained for code validation. More experimental details can be found in the work of Saxena and Prasser (2016). Details of the flow geometry and conditions are mentioned in Table 1. The flow rates were selected to limit liquid entrainment by the turbulent gas flow thereby focusing on simplified flow dynamics.

2. Numerical method

The different turbulence modeling strategies for unsteady flow simulations make use of the same basic form of the governing equations through either a time or space-averaging operation of the Navier–Stokes equations. Even though LES differs from RANS modeling in its ability to capture the effects of large-scale motions in the flow, the small-scale dissipative flow phenomenon needs to be modeled. With respect to these turbulence models only a few details are mentioned here while the readers are referred to the works of Moeng (1984) for theoretical details and Labourasse et al. (2007) for applications of LES in two-phase flows, and of Durbin (2002) for URANS.

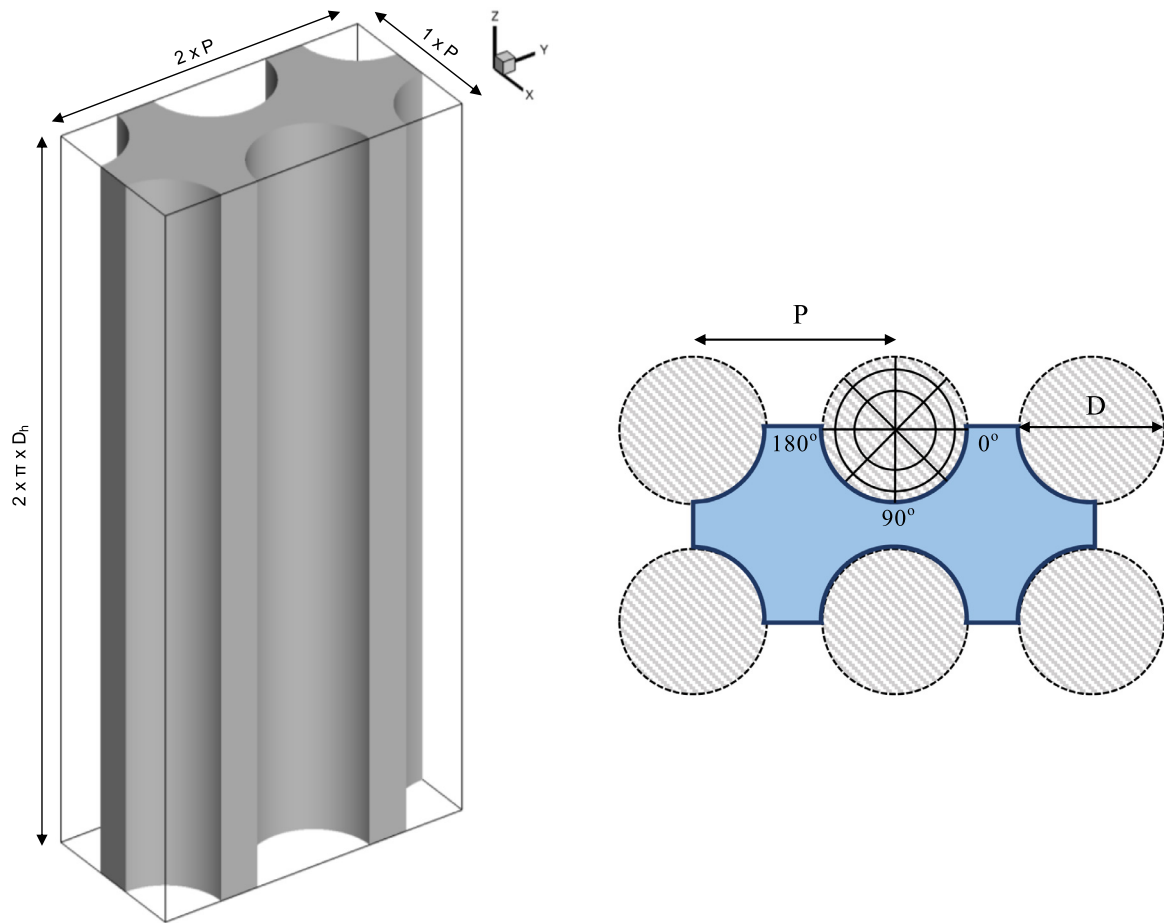


Fig. 2. Geometry of computational domain.

2.1. Physics models

2.1.1. Turbulence models

LES is a turbulence modeling approach based on spatial filtering of the Navier–Stokes equations. It allows the resolution of larger eddies on the grid level while modeling the effects of smaller ones that have a more isotropic nature. The LES approach is effective when used for predicting mass, momentum and heat transfer in flows where the transport processes of interest are dominated by the resolved, large-scale motions with a subsequent cascade of energy to the statistically isotropic and universal small scales. On the contrary, in applications where the controlling processes occur at these smallest scales, such an approach might be less appropriate.

For near-wall flows at high Reynolds numbers, an important phenomenon is the momentum transfer due to the interaction of the flow of the outer layers with the boundary layer flow. While these motions can be resolved by LES, the flow in the region close to the wall is comprised of small length scale viscosity-dominated structures. In order to avoid huge computational costs required to resolve this near wall region, models must be employed to capture the relevant flow phenomena close to such boundaries. By virtue of the large density differences between the gas and liquids in annular flow in practical applications, the gas flow near the interface may be seen as analogous to turbulent flow close to a wall. Since the small-scale motions in the gas flow may be coupled to and associated with localized small-scale interfacial structures, they are not considered to be influential in determining global time-averaged liquid film thickness (LFT) distribution or tracer mixing within the liquid film phase. That is, only large-scale

eddies in the gas-phase are suspected to have a considerable effect on the mixing process within the liquid film through the transport by large-scale interfacial structures. Hence, the LES approach is justified for studying two-phase annular flow in the current context. If, on the other hand, one were to study mass or species-transfer across the gas-liquid interface, the rate-controlling processes would be expected to be largely driven by the small-scale motions. Therefore, the accurate resolution of the small-scale flow motion would be indispensable in such a case. For such an analysis, LES predictions have a first-order dependence on these models. RANS models, on the other hand, are better suited to modeling unsteadiness in the flow due to geometric heterogeneity rather than in stationary flow.

In channels of non-uniform flow cross-sectional area, secondary flow structures of Prandtl's second kind may develop (Bradshaw, 1987). Linear eddy viscosity models assume isotropy for the Reynolds stress components, an assumption which may not hold true for flows with varying boundary layer thickness. Work by Ikeno and Kajishima (2010) has shown that the turbulence energy transfer in the rod gap is suppressed in the wall-normal direction and released in the wall-tangential direction. Models that take into account the anisotropy between the normal turbulent-stress components are known to help better predict flows in which secondary flow structures exist. See for example Benhamadouche (2017), Vitillo et al. (2015), and Baglietto (2006). To account for anisotropic effects, a non-linear SST $k-\omega$ turbulence model with a quadratic constitutive relation (QCR) between the strain and vorticity-rate tensors is tested and compared against a two-layer realizable $k-\varepsilon$ model with a linear eddy-viscosity correlation. The relation of

Table 2
Comparison of flow parameters.

	Temp (°C)	Pressure (MPa)	Rod diameter (<i>D</i>) (mm)	Hydraulic diameter (<i>D_h</i>) (mm)	Mass flux (kg/m ² s)	Outlet steam quality/gas mass fraction
BWR (in operation)	~285	7.6	~12.3	~15 (central subch.)	~3000	~0.15
Experiments (Damsohn and Prasser, 2010; Saxena et al., 2016)	~25	0.1	20	18.85	~40–235	~0.15–0.54

Table 3
Details of computational mesh.

	Cell count	Δz (mm)	$\Delta \theta_{avg}$ (°)	Δr_{min} (mm)	Δt (μs)
Realizable $k-\varepsilon$	14,980,000	0.237	1.098	0.006	20
SST $k-\omega$, QCR	14,980,000	0.237	1.098	0.006	20
LES	28,100,000	0.237	0.720	0.005	5

the Reynolds stress tensor, $\mathbf{T}$, to the strain rate tensor, $\mathbf{S}$, and the vorticity tensor, $\mathbf{W}$, when using the QCR non-linear eddy viscosity model is given in STARCCM+ as follows:

$$\mathbf{T} = 0.0929 \times \mu_t (\mathbf{S} \cdot \mathbf{W} - \mathbf{W} \cdot \mathbf{S})$$

To model the effects of the subgrid scale turbulence, the Wall Adapting Local-Eddy (WALE) model (Nicoud and Ducros, 1999) was used in the framework of LES. A default value of 0.9 was used as the turbulent Schmidt number with all the turbulent models.

2.1.2. Interface-tracking model

The VOF model (Hirt and Nichols, 1981) is an interface-tracking method in which a single velocity field represents the multiple phases while the volume fraction field is solved using a convection equation:

$$\frac{\partial \Phi_i}{\partial t} + \mathbf{u} \cdot \nabla \Phi_i = 0 \quad (1)$$

where $\mathbf{u}$ is the flow velocity field. The volume fraction field is then geometrically reconstructed to obtain the interface between the various phases.

The default HRIC scheme (Muzaferija, 1998), which provides a smooth blending of HRIC in low CFL (Courant–Friedrichs–Lewy) regions and an upwind scheme in high CFL regions of the flow, was used for this work. The final scheme includes factors for interface sharpening applicable in the case of excessive numerical diffusion. A good result was achieved when it was increased to 0.1 for LES. The interface was identified as the iso-surface of the liquid volume fraction variable with a value of 0.5. A small time-step (in Table 2) helped to limit $CFL < 1$ close to the interface.

2.2. Mesh

A high density mesh close to the walls was generated to resolve the dynamics of the liquid film. To achieve this, a structured mesh with fine prism layers around the fuel rod surfaces was employed. Horváth and Dressel (2012) have previously tested the influence of the structure of the mesh on single-phase flow in a square-lattice bundle geometry and found that better results are obtained with structured, body-fitted grids. The mesh distribution across a cross-sectional plane of the geometry used for the LES is shown in Fig. 3 while the cell sizes are listed in Table 3. The minimum cell thickness in the radial direction is found at the wall.

A measure of the grid quality for an LES study is the ratio of the sub-grid scale turbulent kinetic energy to the total turbulent kinetic energy. This parameter can be estimated through the SGS turbulent kinetic energy function (SGSTurbulentKineticEnergy) which is made available by turning on the aeroacoustics model

along with the LES model in STAR-CCM+. The ratio is formulated as

$$M = \frac{k_{sgs}}{k_{sgs} + k_{res}} \quad (2)$$

where k_{sgs} is the SGS turbulent kinetic energy and k_{res} is the resolved turbulent kinetic energy. The resolved turbulent kinetic energy in the LES can be estimated by:

$$k_{res} = \frac{1}{2} (\tilde{u}_i - \langle \tilde{u}_i \rangle_T) (\tilde{u}_i - \langle \tilde{u}_i \rangle_T) \quad (3)$$

where $\langle \tilde{u}_i \rangle_T$ is the time averaging operator over the velocity field given by $\tilde{u}_i$. A value close to zero indicates a refinement appropriate for DNS studies whereas a value of one is acceptable only for RANS modeling. The instantaneous and time-averaged values of this ratio are shown in Fig. 4. Although some of the instantaneous values close to the interface were over 0.15 on the gas-phase side, the time-averaged values were well below 0.1, confirming the quality of the grid.

2.3. Inlet conditions

The generation of fully turbulent inlet conditions is necessary for performing unsteady simulations. The superposition of pseudo-random coherent fluid motions to mean values is important in obtaining the correct growth and sustenance of turbulence in the computational domain for an LES study. This is borne out of the fact that the grid resolved structures always include a time-varying component, which need to be taken into account at the inlet by some mechanism to generate stochastic fluctuations that approximate 'turbulence'. Allowing the flow to develop spatially in a channel spanning several hundred hydraulic diameters in length is too computationally expensive in most cases. As the experimental data was obtained under developed flow conditions, a periodic computational flow domain was deemed necessary to have a valid one-to-one comparison.

Several techniques can be used to generate inflow turbulence for LES studies. Keating et al. (2004) have classified these techniques into three types, i.e., the recycling methods, the precursor databases, and the synthetic turbulence methods. For the former two kinds of generation methods, auxiliary computational domains, which are isolated or connected with the main domain, are needed. As these methods utilize the same flow geometry as the main case, the turbulent structures develop naturally inside the flow domain itself. The synthetic turbulence methods work on the principle of superposing random fluctuations with given moments and spectra on to the mean velocity profile. Their development is still an active field of research (Smirnov et al., 2001; Battean et al., 2004). The synthetic eddy method (SEM), proposed by Jarrin et al. (2006), is the in-built method in STAR-CCM+.

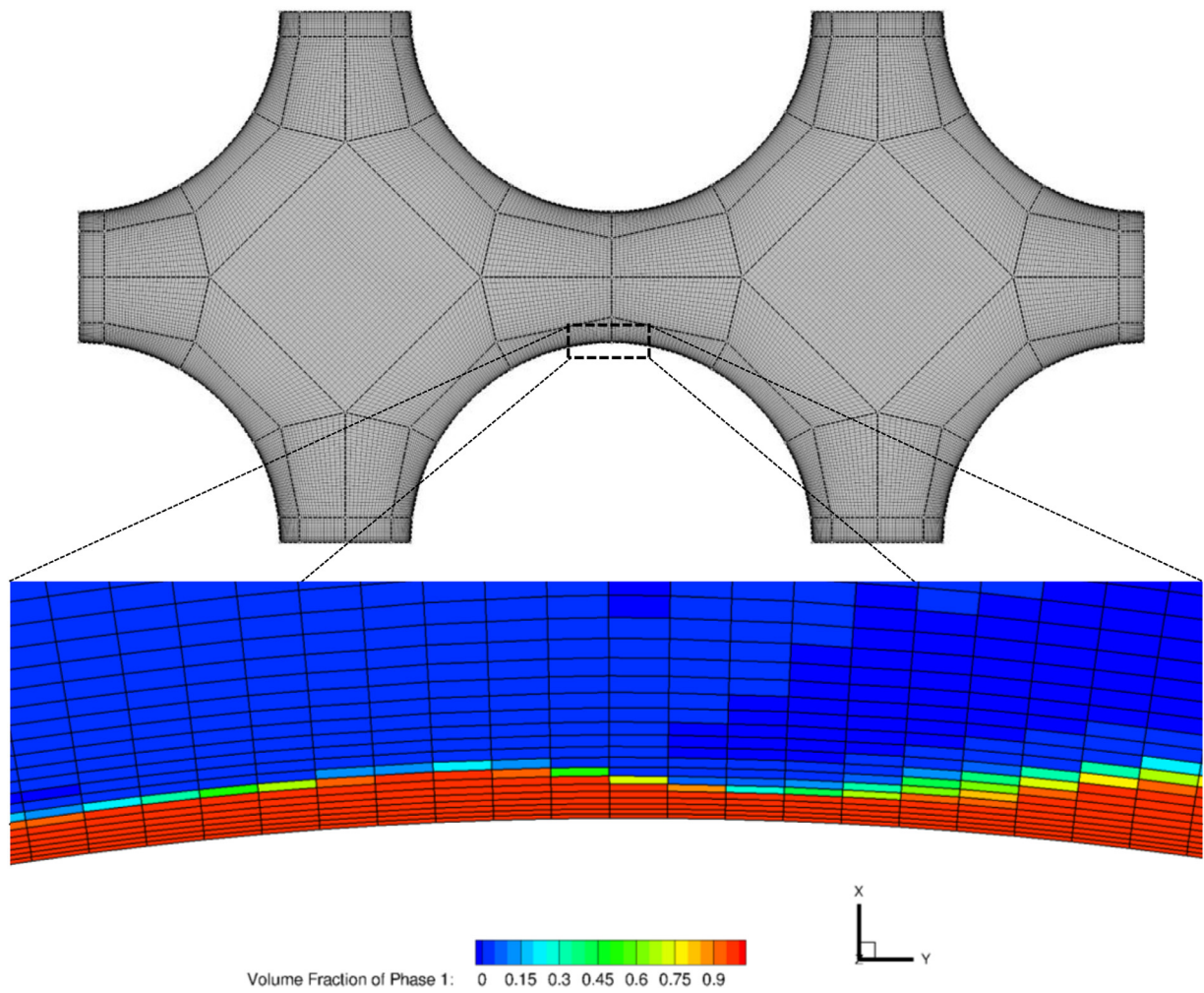


Fig. 3. Computational mesh used for LES.

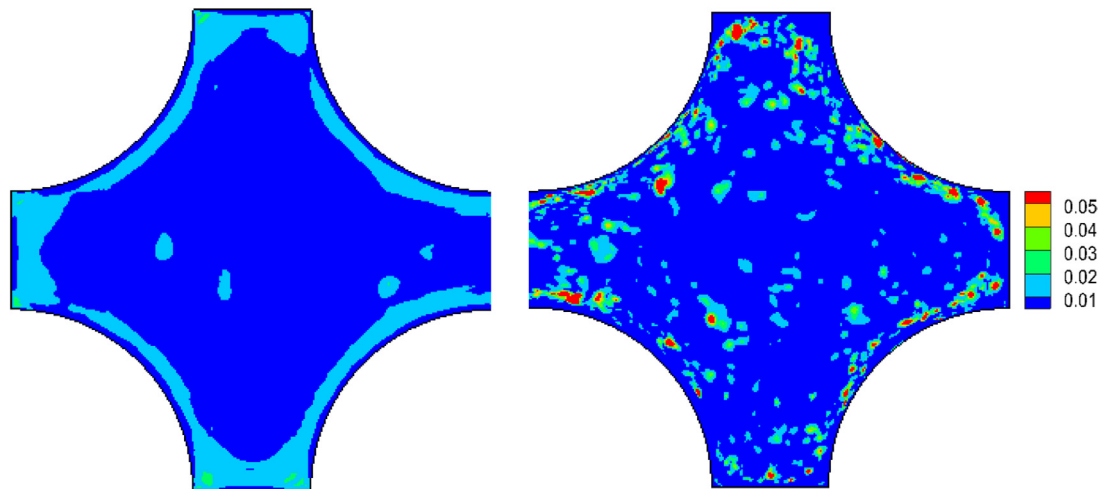


Fig. 4. Time-averaged (left) and instantaneous (right) distribution of M ratio.

The precursor simulation method makes use of a separate initial simulation of the flow to generate turbulent inlet conditions. The result of the precursor simulations is stored to be used later for the main computation. The generation of precursor databases is often cumbersome due to the large memory re-

quirements. Also, periodic recycling conditions are also employed for precursor simulations. Since such a database is available only for a finite time-period, it may introduce periodicity into the turbulent inlet data leading to an unrealistic periodicity in the flow itself.

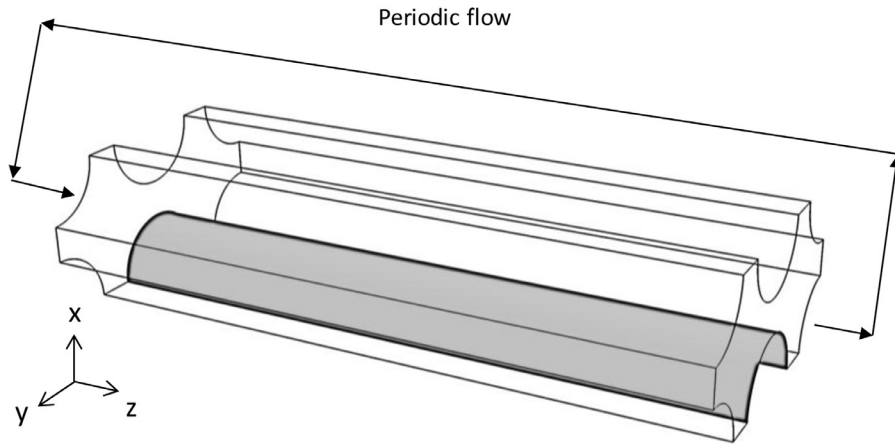


Fig. 5. Periodic flow domain for numerical computation.

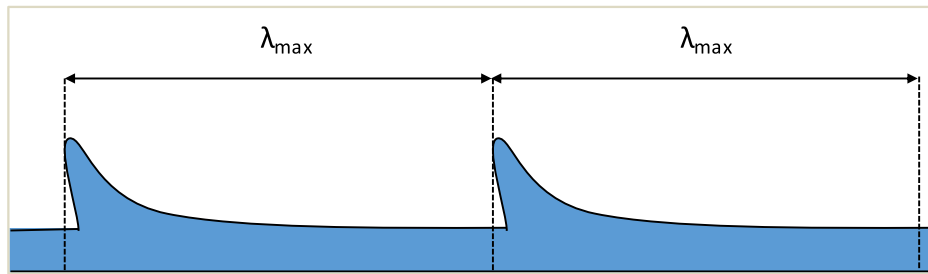


Fig. 6. Characteristic length scale in flow direction.

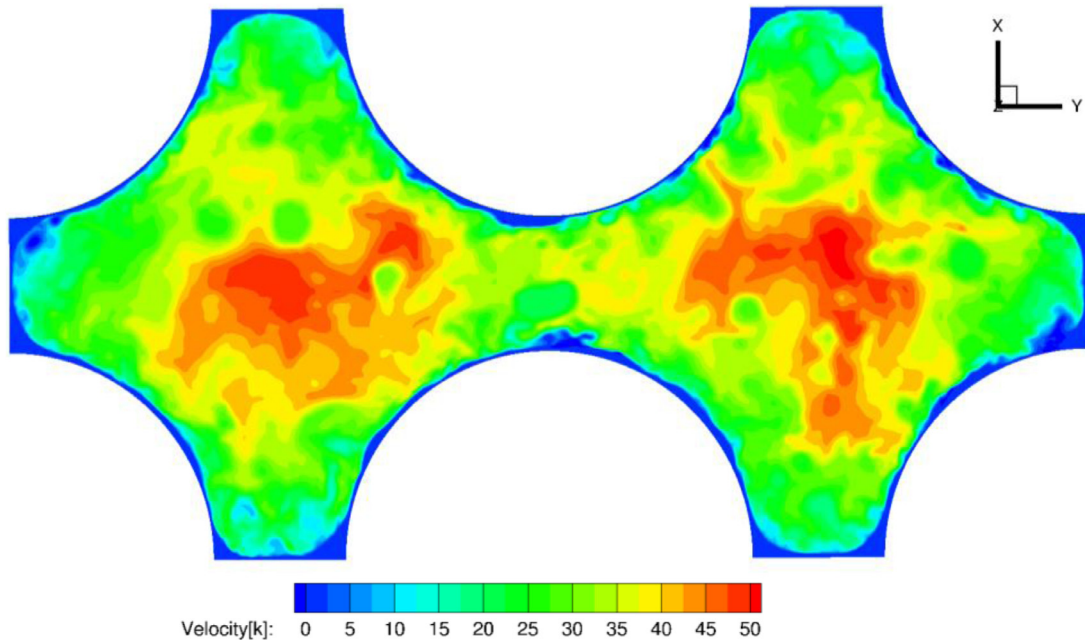


Fig. 7. Instantaneous streamwise velocity field with the recycling method (LES).

2.4. The recycling flow method

The simplest method for generating turbulent inlet conditions is based on recycling the data from the outlet to the inlet of the calculation domain (see Fig. 5). A direct mapping of the flow field from the outlet to the inlet can result in a slightly different mass flux to be carried over due to error propagation arising from solver

convergence. A slightly different method of tackling the problem is by defining a mean pressure gradient in the streamwise direction. This allows the flow to develop with the phasic mass flow rates associated with the applied mean pressure gradient. If this mean pressure gradient is not known *a priori*, an iterating procedure to estimate it is then required to recover the phasic mass flow rates. To avoid this tedious procedure, a method based on the internal

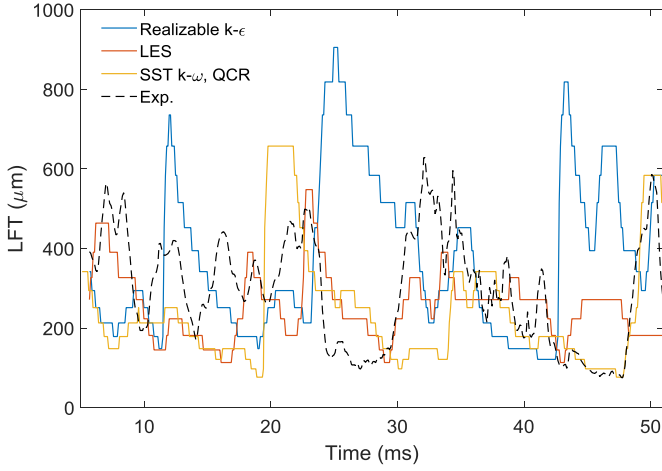


Fig. 8. Asynchronous LFT time series, experiments vs. CFD ($\theta = 90^\circ$).

mapping method of [Baba-Ahmadi and Tabor \(2009\)](#) has been developed for two-phase flows as part of this work.

The method requires the knowledge of only the mass flow rates of the individual phases. Before outlining the method developed for two-phase flows, the existing method for single-phase flow is described. The mass conservation principle implies a constant mass flow rate for an incompressible single-phase flow through a conduit:

$$\dot{m}_0 = \rho \cdot A_{ch} \cdot u_n = \text{const}$$

The area of the channel A_{ch} being constant, the outlet velocity field can then be mapped back to the inlet through a scaling factor

$$u_{in} = (\dot{m}_0 / \dot{m}_{out}) \cdot u_{out} \quad (4)$$

where u_{in} is the mapped inlet velocity field using the outlet velocity u_{out} from the previous time-step. Only velocities normal to the inlet and outlet are used in this step. The rescaling factor limits and corrects the propagation of numerical errors in the time evolution of the flow to maintain a constant mass flow rate $\dot{m}_0$. For two-phase flow, both the velocity field and the phasic volume fraction variables are required to be recycled. The behavior of two-phase flow is different from single-phase flow in another aspect.

The mass flow rate of each phase across any particular flow cross-section varies with time due to the presence of unsteady flow features like waves, etc. Hence, the given equality doesn't hold any longer:

$$\dot{m}_\phi = \rho_\phi \cdot A_\phi \cdot u_{n_\phi} \neq \text{const}$$

As a result, the same rescaling method cannot be applied for a two-phase flow using the instantaneous velocity field at the out-flow plane as in [Eq. \(4\)](#). Instead, the phasic mass flow rates averaged over the time-period of the longest wavelength can be considered constant. Since the use of a time-averaging operator is not suitable for providing time-resolved turbulent inlet conditions, the space-averaging operator may instead be used. In essence, one can substitute the time averaging operation over the time-period of the longest wavelength at a certain cross-section with a spatial averaging operation of the flow over the entire wavelength at some particular instant of time. In mathematical terms, this is described by the ergodic assumption, which states the equivalence of spatial and time averages of a stationary homogeneous turbulent flow.

$$\frac{1}{T} \int_t^{t+T} Q(t^*) dt^* \equiv \frac{1}{\lambda_{max}} \int_x^{x+\lambda_{max}} Q(x^*) dx^*$$

The choice of the averaging length λ_{max} is correlated to the integral length scale of the computational domain (given by $2\pi \cdot D_h$ where D_h is the hydraulic diameter) or the largest wavelength ([Fig. 6](#)).

For a given superficial velocity J_ϕ for phase ϕ and channel cross-sectional area A_{ch} , the integral phasic mass flow rate in the domain is given by:

$$\begin{aligned} \int_0^V \dot{m}_{\phi_0} dV &= \rho_\phi \cdot J_\phi \cdot A_{ch} \cdot \lambda_{max} \\ \int_0^V \dot{m}_\phi dV &= \rho_\phi \cdot \sum_V [A_{cell} \cdot h_{cell} \cdot u_n] \cdot \Phi_i \end{aligned} \quad (5)$$

where $A_{cell} \cdot h_{cell}$ is the volume of the cell, u_n the streamwise velocity component, Φ_i the phasic volume fraction and the summation of the bracketed quantity over the complete volume V of the domain. The quantity $\dot{m}_{\phi_0}$ represents the target phasic mass flow rate and can be determined prior to the simulation from the density and superficial velocity of each phase, and the geometric characteristics of the flow domain, namely the cross-sectional area and the axial flow length. The mapped inlet velocity from the outlet

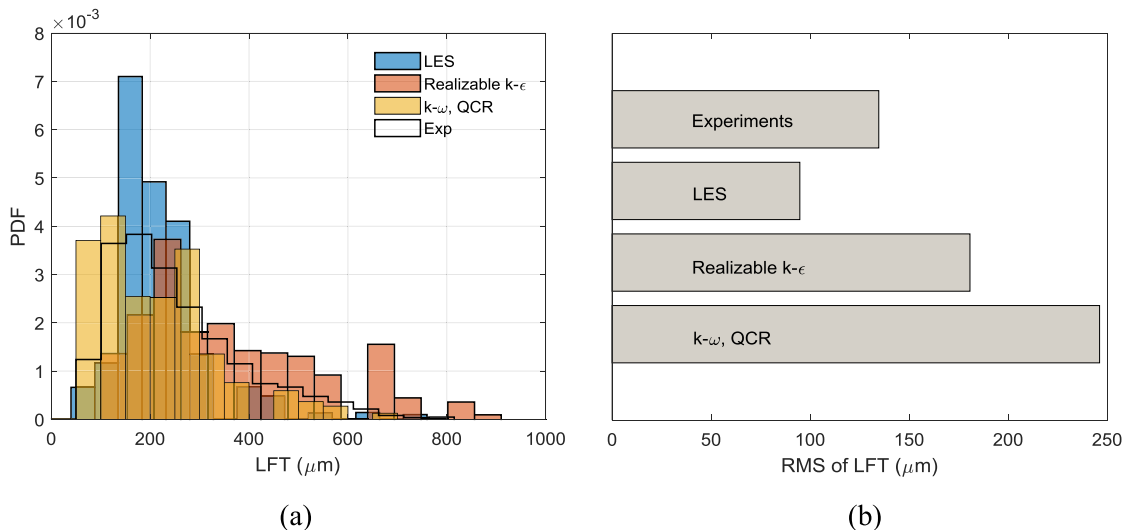


Fig. 9. (a) LFT probability distribution and (b) RMS of LFT at point probe ($\theta = 90^\circ$).

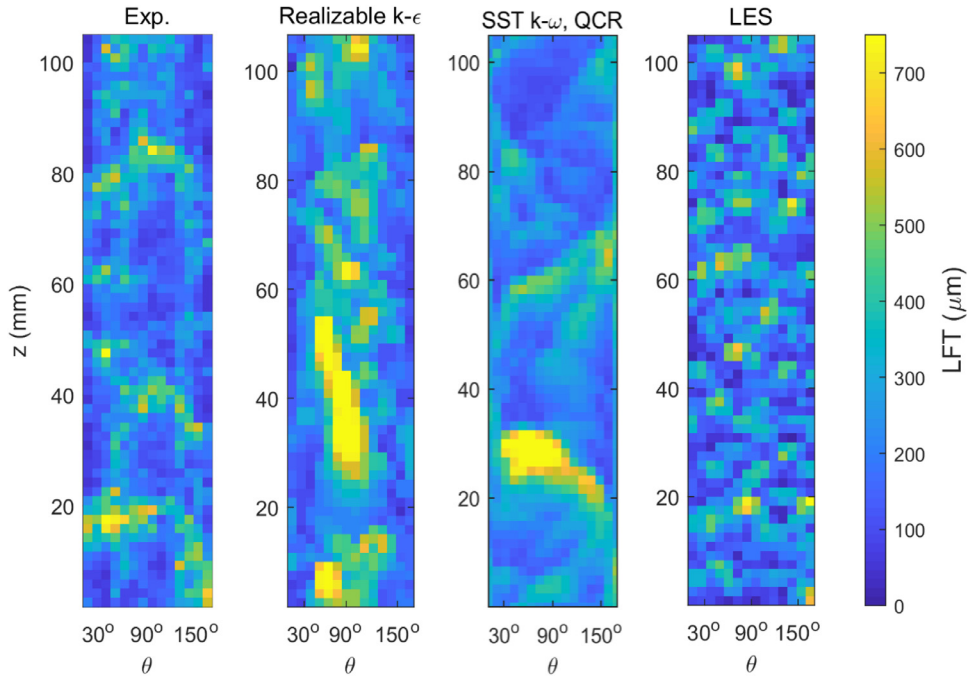


Fig. 10. Instantaneous LFT distribution, Experiments vs. Two-phase CFD (64×16 grid).

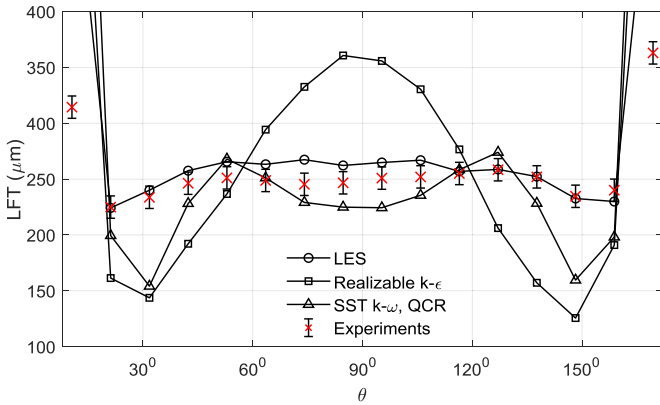


Fig. 11. Time-averaged LFT around half rod.

plane then becomes:

$$u_{\phi_{in}} = \left(\int_0^V \dot{m}_{\phi_o} dV / \int_0^V \dot{m}_{\phi} dV \right) \cdot u_{\phi_{out}} \quad (6)$$

The integrals in Eq. (6) can be easily calculated in any commercial code, for example using field functions and reports in STAR-CCM+. The developed velocity field, obtained through the natural development of the turbulent structures with the recycling method, is shown in Fig. 7. The high turbulent intensities and strain rates may cause the SEM to generate a large number of small eddies with high vorticities close to the gas-liquid interface. Through the recycling flow method, concurrent distributions of the velocity field and phase distributions are simultaneously made available at the inlet, which is not possible with the SEM method. An initial film of $500 \mu\text{m}$ thickness was set as the starting condition on all the rods, the rest being gas in the channel core. The phasic velocities were then estimated by the given superficial velocities and the flow area occupied by the two phases.

3. Results and discussion

The flow was initially allowed to develop for 10 flow through times based on the mean liquid film velocity given at the inlet (1.2 m/s). The necessary condition for flow development was determined by the asymptotic behavior of the turbulent kinetic energy (TKE) monitor averaged over the flow domain. Flow statistics were collected over an additional 10 flow through times.

3.1. Liquid film characteristics

The distribution of the liquid film over a fuel rod is discussed in this section. Several researchers, e.g. Chien and Ibele (1964) and Ueda and Nose (1974), have found experimental evidence of a two-layer flow structure in liquid films. This flow structure has been widely utilized in the modeling of annular flow since its proposal, for example by Dobran (1983). The continuous layer close to the wall is estimated to be relatively thin and associated with a low heat transfer coefficient. A wavy layer, which is intermittent in nature, is present on top of this base layer. The waves are known to travel at much higher velocities, carrying a significant part of the liquid mass flow in the process. The regions under the disturbance waves have been experimentally found to have preferential nucleate boiling sites by Barbosa et al. (2002).

An asynchronous LFT time series extracted at the subchannel gap position ($\theta = 90^\circ$) is shown in Fig. 8. The dynamics of the wavy layer were found to be dependent on the turbulence model used. Probability density functions (PDFs) of LFT comparing different turbulence models are shown in Fig. 9(a). To measure the intensity of LFT fluctuations, the RMS of the film thickness was estimated at the subchannel gap position (90°). The LES predicted a high-peak low-variance distribution around the mean film thickness value due to the absence of any large-amplitude wavy structures. The realizable $k-\epsilon$ model predicted a thicker wavy layer compared to the experiments with a peak around $700 \mu\text{m}$. The largest RMS of the film thickness was obtained with the non-linear $k-\omega$ model (see Fig. 9(b)).

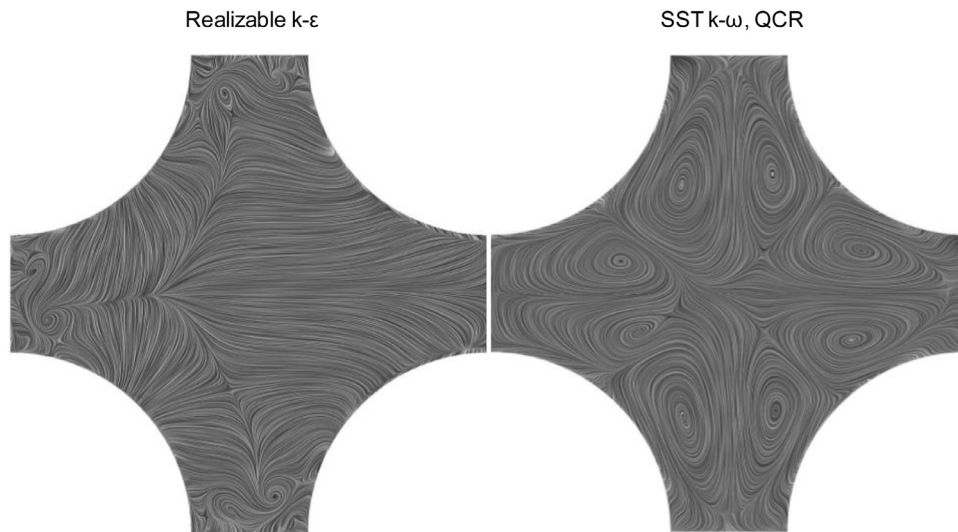


Fig. 12. Secondary flow structures from mean flow.

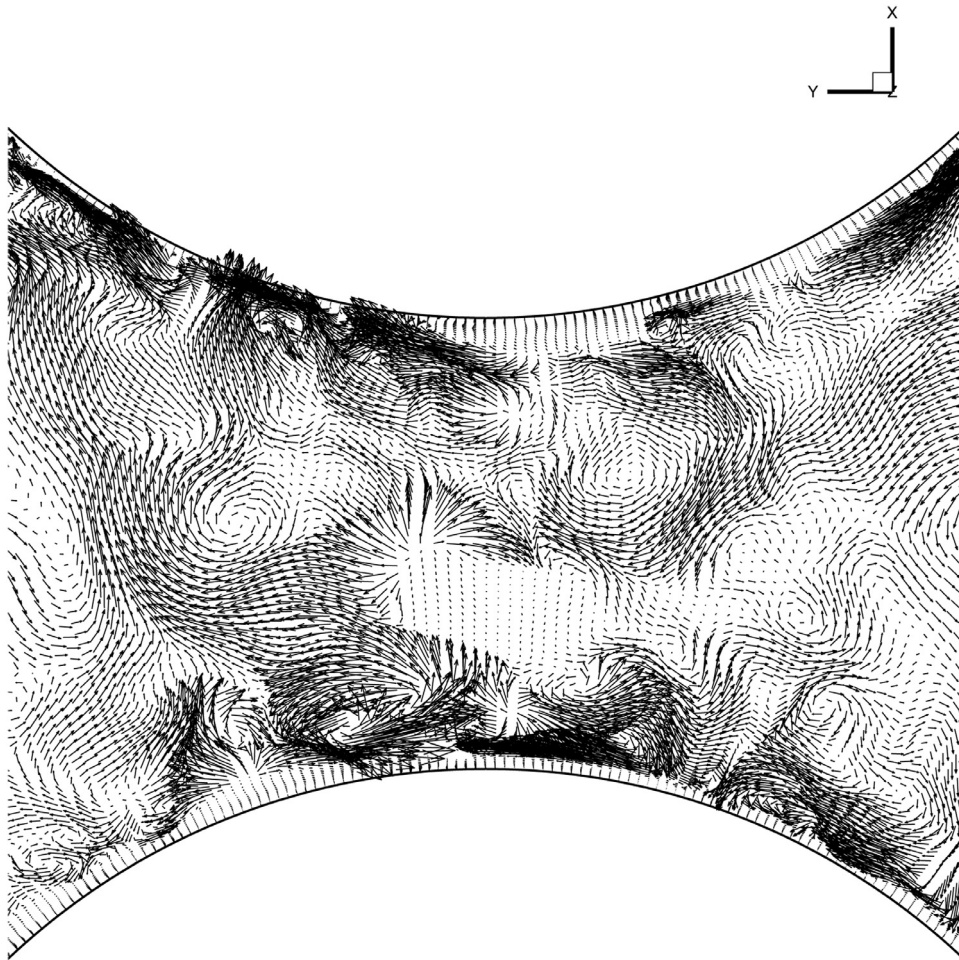


Fig. 13. Tangential velocity vectors in subchannel gap region (LES).

Snapshots of the unsteady gas-liquid interface obtained with the different turbulence models around a half fuel rod, resolved on a 64×16 mesh, are showed in Fig. 10 and 12. Disturbance waves may exist in a stable state for several wavelengths in the flow direction. These were distinguishable in the experiments as packets of thick film and recorded to be asymmetric about the fuel rod cir-

cumference (Damsohn and Prasser, 2010). The blockage of the side subchannel gaps is not deemed to be the cause of this observation due to the symmetry about the central subchannel gap. Although film structures with high LFTs were predicted by the realizable $k-\varepsilon$ model, they were observed to have a low circumferential coherence. The SST $k-\omega$ non-linear eddy viscosity model predicted tilted

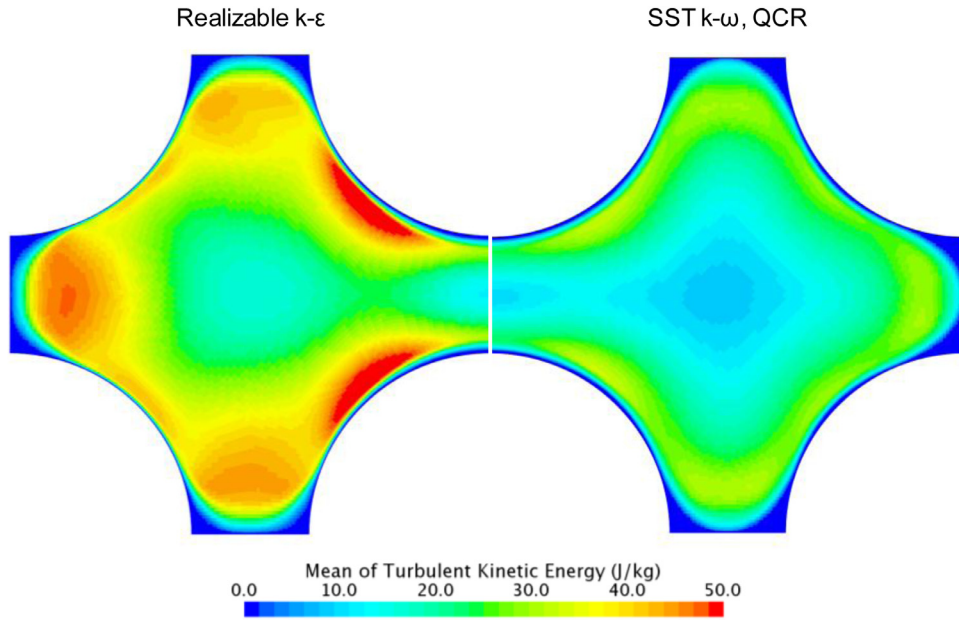


Fig. 14. TKE distribution from two different URANS models.

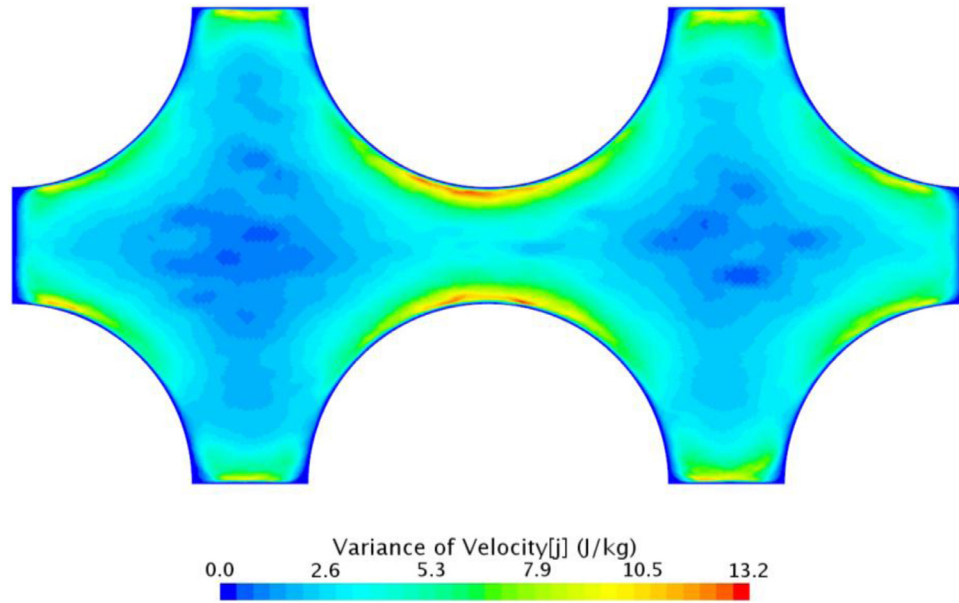


Fig. 15. Variance of cross-velocity (u_j) from LES.

or oblique wavy structures, similar to those observed in the experiments. The sharpening coefficient used for the VOF discretization scheme with the LES model may prevent the generation of coherent roll waves from localized ripple waves, leading to the direct dispersion of the liquid phase as droplets in the gas phase instead. Moreover, some two-phase turbulence effects may become significant on approaching the interface due to the large density gradient. For example, a model to account for turbulence damping near the gas-liquid interface, currently absent in commercial codes with LES, may help to sustain such waves in the flow. Based on these results, it is recommended to study these effects on the flow in future work.

Time and space-averaged (in streamwise direction) LFT distribution profiles across the span of a half fuel rod are shown in Fig. 11. A quantitative comparison of the performance of different models is shown in Table 4. The mean LFT was calculated by ignoring the

two grid points next to the corners. A good agreement was found between LES, the SST $k-\omega$ model and the experiments. Although the realizable $k-\epsilon$ model predicted the mean LFT to an accuracy of 2%, the spanwise distribution revealed a peaked distribution in the subchannel gap region with a large deviation from the experimental distribution. The non-linear SST $k-\omega$ model gave good predictions for the mean LFT far away from the rod ends and reproduced the peaks observed at around 55° and 125° polar positions on the rod. The secondary flow structures (Fig. 12) were found to affect the lateral distribution of the LFT on fuel rods through cross-channel mass transport. The improved predictive performance of the non-linear eddy viscosity correlation in the SST $k-\omega$ model is linked directly to its ability to resolve the secondary flow structures.

Inter-channel exchange of momentum and energy by large-scale coherent structures is well documented for single-phase flows, for

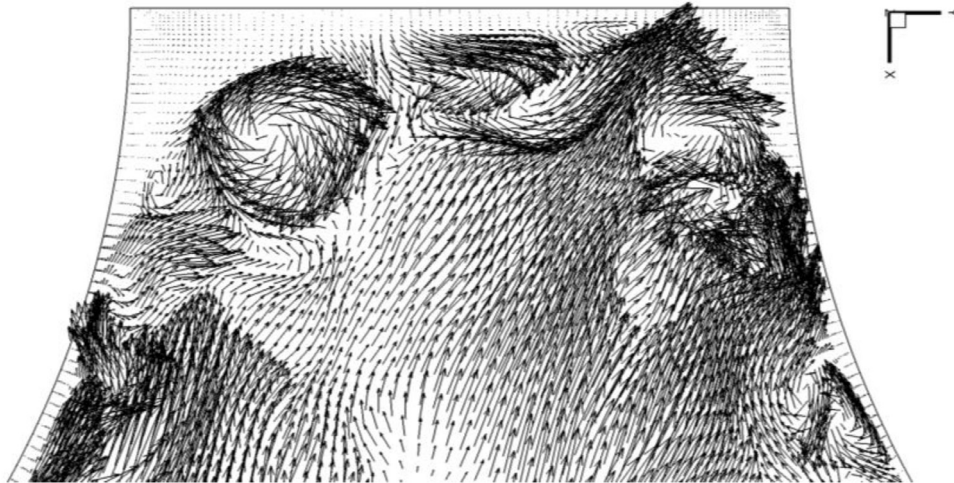


Fig. 16. Planar-velocity vector field near corners (LES).

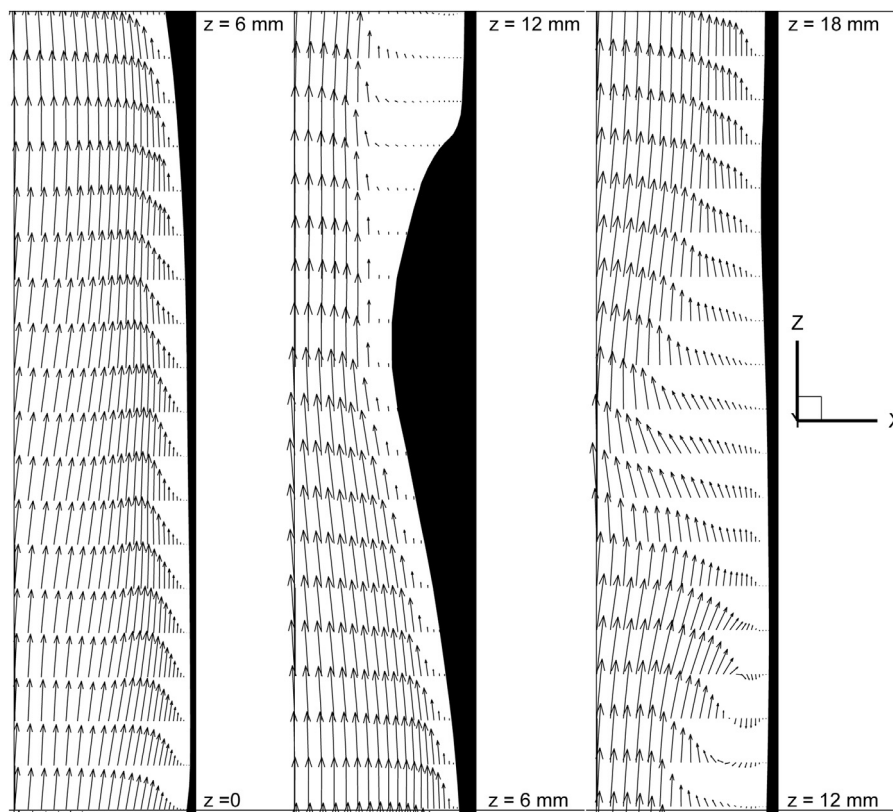


Fig. 17. Velocity vectors in gas flow above wave (LES).

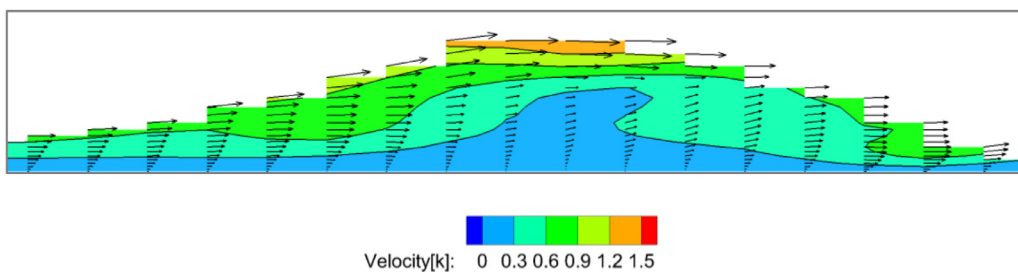


Fig. 18. Velocity vectors in liquid wave (LES).

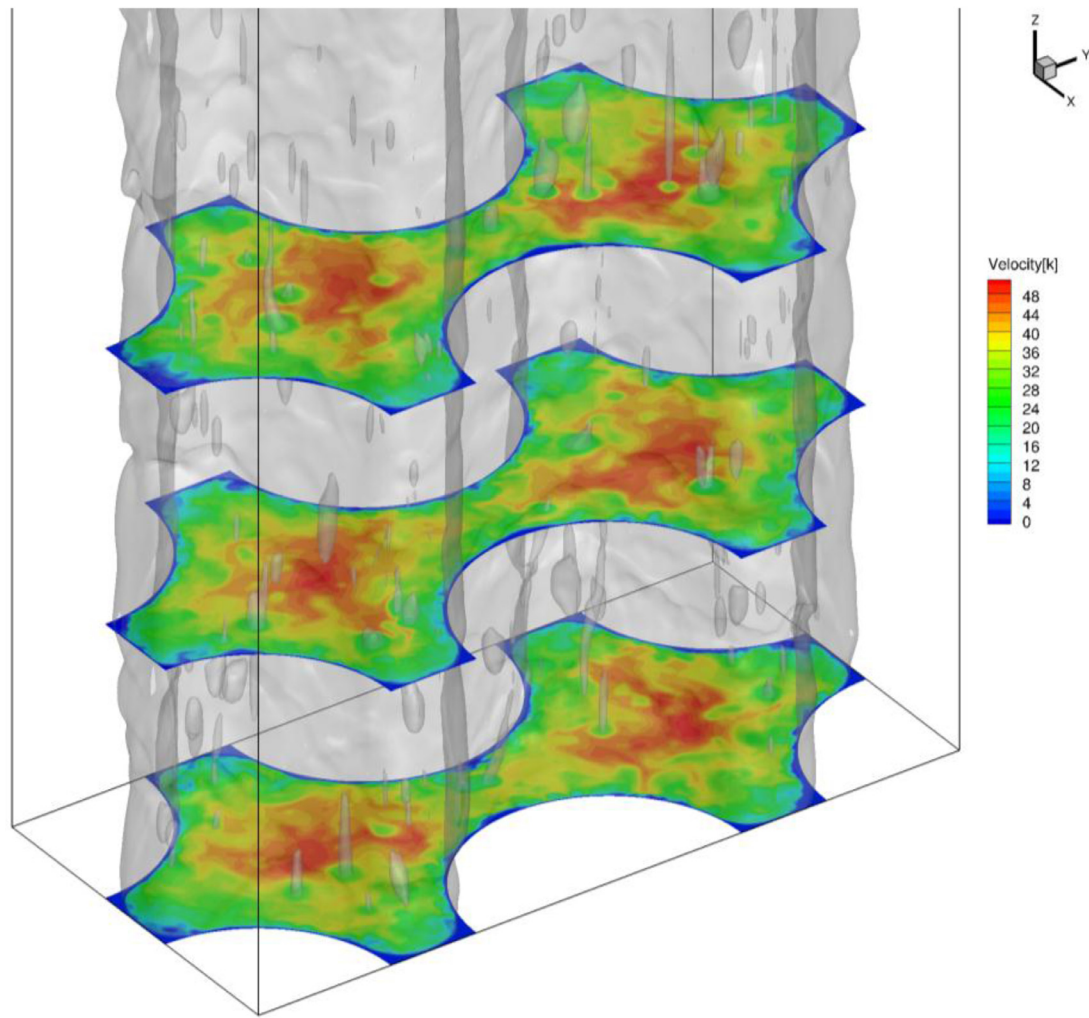


Fig. 19. Instantaneous snapshot of streamwise velocity field with highlighted gas-liquid interface (LES).

Table 4
Comparison of mean LFT.

	Mean LFT (μm)	Error, mean LFT (%)	Maximum pointwise error, LFT (%)	θ , Maximum error ($^\circ$)
Experiment	245	0	0	
LES	253	3.3	9.0	74
Realizable $k-\varepsilon$	240	2.0	46.2	85
SST $k-\omega$, QCR	224	8.6	6.7	53

example by Rehme (1992). Their existence in the gas-core of annular two-phase flow and their influence on the liquid film covering the fuel rods remains unknown due to the difficulty in obtaining such evidence. These periodic structures can suppress or delay CHF by increasing turbulent mixing and preventing the formation of local hot spots on fuel rods. Local eddies in the spanwise direction were predicted by LES in the subchannel gap region (see Fig. 13) although no streamwise coherence was detected between them. Increased turbulent activity was recorded in the subchannel gap by the LES but was absent in the URANS predictions. It should be noted that for large P/D ratios like in the current work ($P/D = 1.325$), strong flow pulsations due to coherent turbulent structures in a subchannel geometry have not been measured (Meyer, 2010).

Time-averaged TKE distribution comparison between the linear ($k-\varepsilon$) and non-linear ($k-\omega$) eddy viscosity model is shown in Fig. 14. The distribution of the variance of the cross-velocity, $(v')^2$,

computed by LES is shown in Fig. 15. In the given computational domain, a significant contribution to the cross-channel flow effects comes from the v' component. Peaks in both the TKE and the variance distribution close to the time-averaged interface on the gas side is due to the dynamic nature of the interface, similar to the behavior near a wall. Higher variance values were observed close to the open subchannel gap compared to the subchannel centers as the eddies formed near the subchannel gap are not constrained by any no-slip boundaries.

The mass transport of the liquid caused by the corner eddies in the gas flow (see Fig. 16) results in the large film thickness recorded at the polar extremities. As mentioned by Damsohn and Prasser (2010), such an accumulation of the liquid in the corners results in a lower LFT on the rods than in a real bundle for the same flow rates. Although this effect may not be relevant to flow in central subchannels, it is nevertheless an important observation for corner subchannels that can be reproduced computationally.

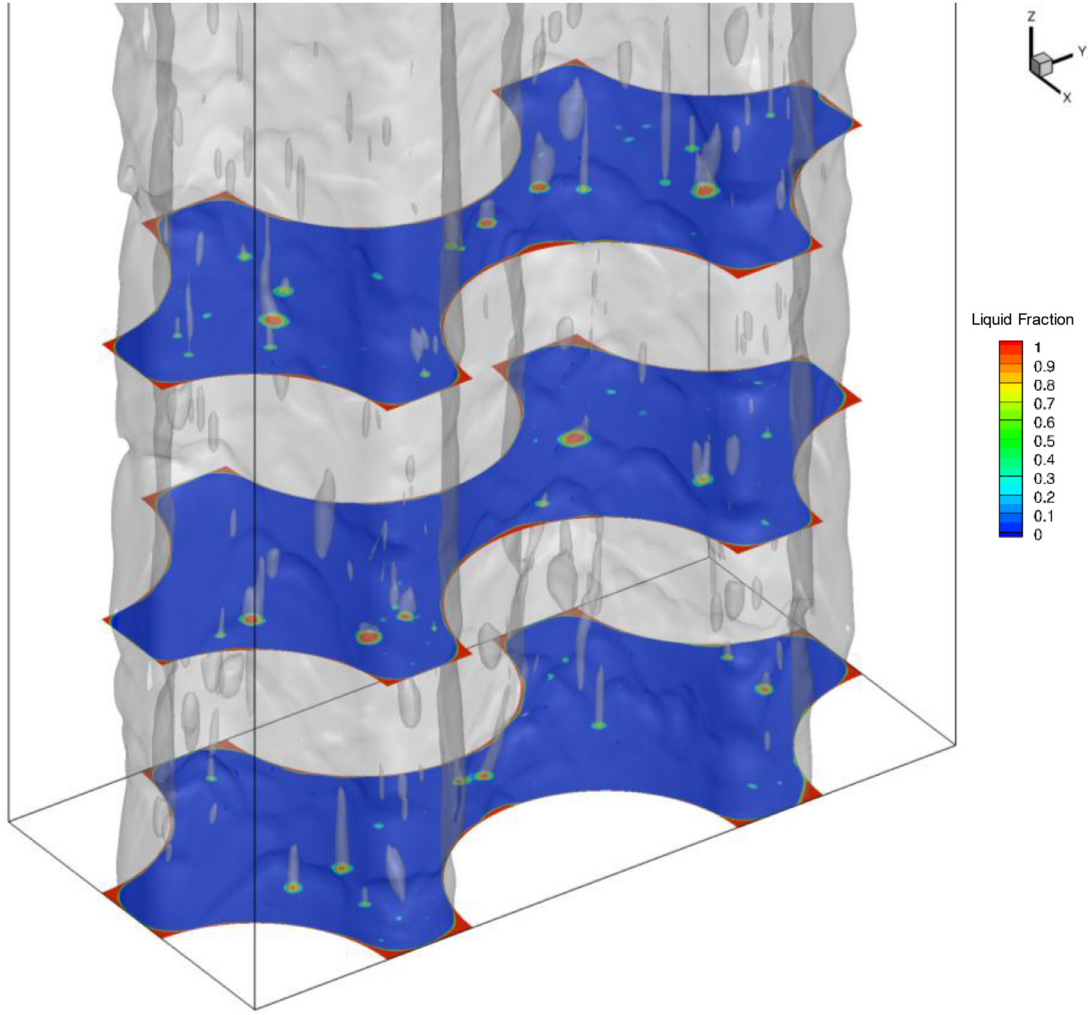


Fig. 20. Instantaneous snapshot of liquid fraction field with highlighted gas-liquid interface (LES).

Perturbation of the gas-phase flow by large wavy structures of the liquid film can result in a coupling of the flow dynamics near the interface. Fig. 17 shows gas velocity vectors over a liquid wavy structure in the subchannel gap vertical cross-section. The gas flow was found to accelerate on approaching the wave crest due to flow-area blockage, eventually resulting in flow separation further downstream. Negative interfacial shear-stresses were predicted downstream of some waves as a result. The pressure differential resulting from the flow separation over a wave leads to a movement of the gas from the core-region towards the liquid film in a direction perpendicular to the interface. This leads to an increased interfacial shear stress on the windward side of the liquid film resulting in the growth of such wavy structures. That is, the asymmetric distribution of shear stresses on the two slopes of a wave is coupled to its growth and sustenance. This may eventually result in the shearing off of liquid from the tips of large waves as has been discussed by Kataoka et al. (1983). A similar wave formation mechanism was also found in the numerical work of Sato and Niceno (2017). These results indicate how the gas-phase boundary layer is highly perturbed by the interfacial features and, hence, needs to be taken into account when modeling annular flow.

The shear and normal stress balance at the interface allows the coupling of the gas-liquid flow in that region. Velocity vectors under a liquid wave are shown in Fig. 18. The liquid close to the interface is accelerated by the higher gas-shear on the windward side while the leeward side gets decelerated. Negative velocities were

found immediately downstream of the waves indicating reversal of liquid flow all the way to the wall.

Due to the nature of the measurement apparatus, no experimental data for the gas-phase was available. The dispersed liquid-phase, captured as elongated spherical droplets in the gas-core by LES, was found to travel at lower flow velocities compared to the surrounding gas phase. These can be seen as distinct circular-shaped regions in the axial velocity field shown for several cross-sections from the LES in Figs. 19 and 20. In the absence of experimental data, no further details on the dispersed droplets were analyzed.

3.2. Velocity profile of liquid film

There exists a broad consensus regarding the flow structure in a turbulent boundary layer for single-phase flows as defined by the log-law proposed by Von Karman (1939), with typical values of the constants $\kappa = 0.4$ and $C = 5.5$. The Universal Velocity Profile (UVP) is formulated as:

$$u^+ = \begin{cases} y^+ & 0 < y^+ < 5 \\ 5 \log y^+ - 3.0 & 5 < y^+ < 30 \\ \frac{1}{\kappa} \log y^+ + C & y^+ > 30 \end{cases} \quad (7)$$

where

$$u^+ = u/u_\tau \text{ with } u_\tau = \sqrt{\tau_w/\rho} \quad y^+ = yu_\tau/\nu$$

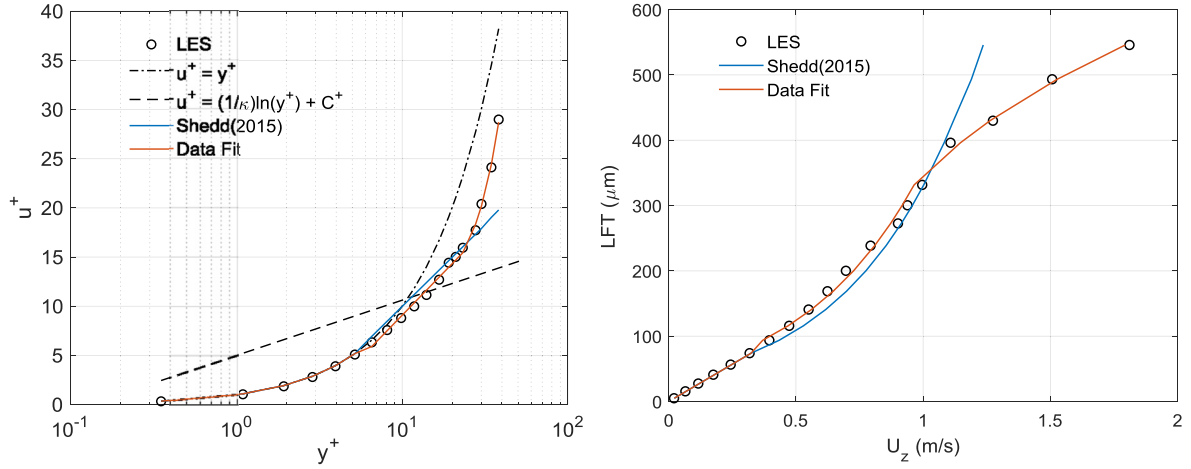


Fig. 21. Time-averaged velocity profile in liquid film in dimensionless (left) and dimensional (right) units.

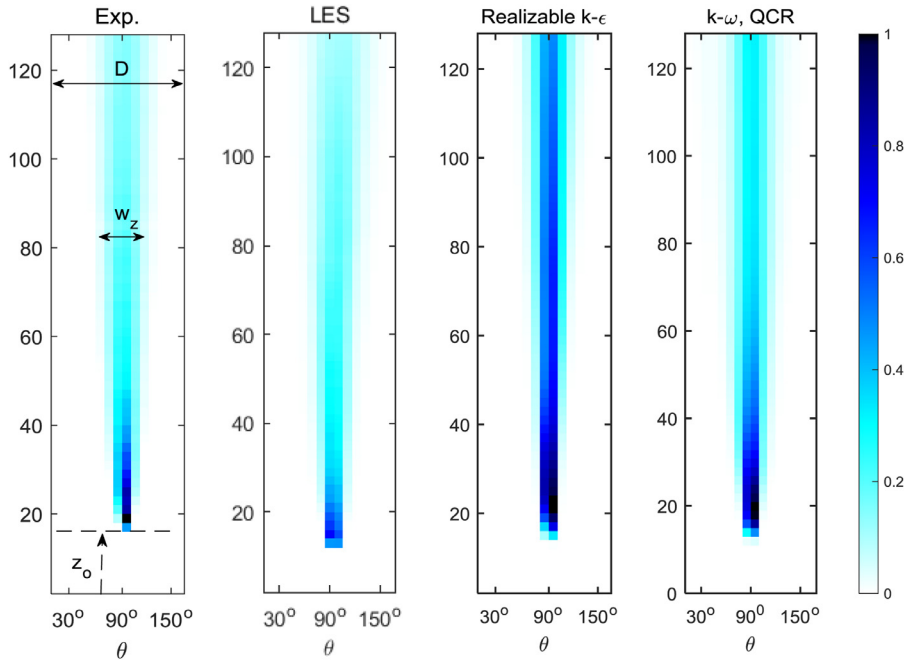


Fig. 22. Time-averaged tracer distribution in liquid film around half-rod (64×16 grid).

It is not known if the classical log-law behavior is also valid for shear-driven liquid films. Due to the action of turbulent gas shear at the interface, the flow structure of a liquid film in an annular flow is thought to be more complex than that of a single-phase, turbulent boundary layer. Measurements performed by Vassallo (1999) confirmed the preservation of the single-phase log-law in thin liquid films. However, for describing thicker liquid films, a modified log-law with a smaller value of C was proposed. It was even proposed to be as low as $C \approx 0.0$ for an air-water annular flow with $J_l = 1.3 \text{ m/s}$, $J_g = 12.8 \text{ m/s}$ and a measured film thickness of 0.69 mm . The velocity profile in the liquid film was found to flatten on approaching the film-gas interface in the work of Vassallo (1999). Liquid film velocity profiles recently obtained by Shedd (2013) through PIV were also found to deviate from the UVP. A modification of the coefficients in the buffer and the log layer was proposed by the authors as:

$$u^+ = \begin{cases} y^+ & 0 < y^+ \leq 5 \\ 7.2 \log y^+ - 6.6 & 5 < y^+ \leq 30 \\ 7.38 \log y^+ - 7.1 & y^+ > 30 \end{cases} \quad (8)$$

This motivated the investigation of the time-averaged flow structure of liquid films from the LES data. The velocity profiles obtained from this work are plotted in Fig. 21. In the laminar sub-layer region, i.e. for $y^+ \leq 5$, the velocity profile was found to obey the linear region of the UVP. Above this layer, the velocities found by LES were larger than those from the UVP but close to those obtained from the profile of Shedd (2013).

The distinct flow structure of liquid films results from the presence of waves that carry a large momentum flux, a feature absent in single-phase flows. The formation of large eddies in a direction normal to the interface is damped due to the surface tension forces acting along the interface. Hence, this decrease in the turbulent mixing length near the interface leads to higher velocities in the liquid film as compared to single-phase flows at a given distance away from the wall. To take this into account, a modified velocity profile for liquid films is proposed here. Considering the effects of the two shear layers, namely the wall and the interfacial shear layer, on the liquid film, an 'S'-shaped velocity profile is proposed here. Since a logarithmic profile is found to fit the so-called buffer layer near the wall, a power profile of the form $a(y^+)^e + b$ is

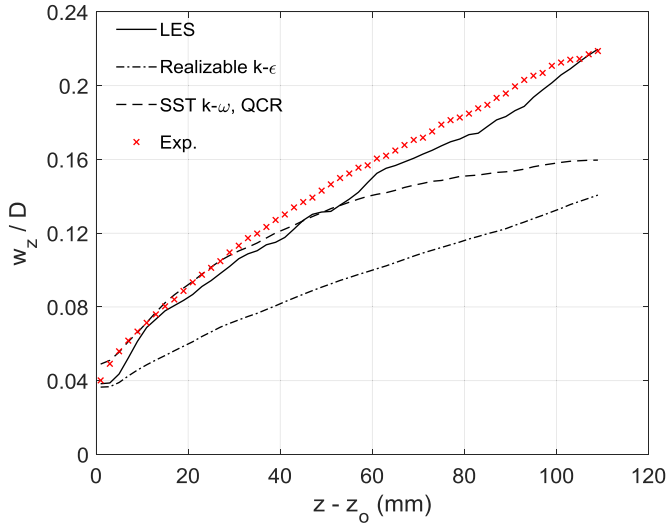


Fig. 23. Tracer mixing width downstream of injection.

chosen to characterize the velocity in the top wavy layer. The following fit for the streamwise velocity profile from the LES data is obtained as:

$$u^+ = \begin{cases} y^+ & 0 < y^+ \leq 5 \\ 7.54 \log y^+ - 8.276 & 5 < y^+ \leq 25 \\ 8.923 \times 10^{-4} (y^+)^{2.718} + 10.89 & y^+ > 25 \end{cases} \quad (9)$$

It is to be noted that the profiles defined here characterize the dependency of the velocity on only the wall distance and no effect of the liquid holdup is included. In other terms, the integration of the proposed velocity profile over the film height will fail to provide accurate mass flow rates for the liquid phase. Instantaneous velocity profiles similar to the 'S'-shaped profile shown in Fig. 21 have been previously reported by Lorencez et al. (1997) in liquid stratified flows. Evidence of eddies under waves has already been mentioned (Hewitt and Jayanti, 1997) and shown in this work (Fig. 18). By Fjorth's theorem this would require an inflection point

in the velocity profile of the liquid film, which the proposed profile satisfies. Since only one set of flow rates are simulated in this work, the dependency of the coefficients a and b on the superficial velocities and fluid properties remains to be studied in future work.

3.3. Tracer mixing

Two-dimensional, time-averaged tracer distributions from the experiments and computations are shown in Fig. 22. The tracer distribution is similar to that characteristic of jets or plumes in terms of the lateral spreading downstream of the injection point. The cross-transport of the tracer is a result of the action of the transverse gas eddies on the liquid film. Time-averaged width of the plume extracted from the experiments and numerical simulations is plotted in Fig. 23. The width was obtained by fitting Gaussian curves over data points on each row of the matrix and is equivalent to the full width half maximum (FWHM) multiplied by a constant (~ 1.18). The limited time statistics of LES data, due to the large computational requirements, result in a slightly non-smooth distribution comparatively.

A good agreement for the mixing width was obtained between the experiments and LES, whereas the realizable $k-\epsilon$ model significantly under predicted it. The SST $k-\omega$ model predicted the mixing width well till 30 mm downstream of the injection point but then starts to deviate. Interestingly, the SST $k-\omega$ model predicts a classic $\sqrt{d}$ behavior of the mixing width as a function of distance. Eddies in the gas flow have a direct effect on the formation of interfacial structures in the liquid film and consequently on the mixing of passive scalar in the liquid film. The under-prediction of the mixing by the $k-\epsilon$ model is supported by the presence of the peak in the LFT distribution at the subchannel gap (Fig. 11). Even though the non-linear viscosity turbulence model is able to reproduce secondary flow in the channel geometry, it is still unable to capture the cross-channel eddies that sustain and enhance mixing far away downstream of the injection point of the tracer. Since the LES approach resolves the inertial subrange scales of the flow, it is able to better predict this cross-channel mixing compared to the URANS models.

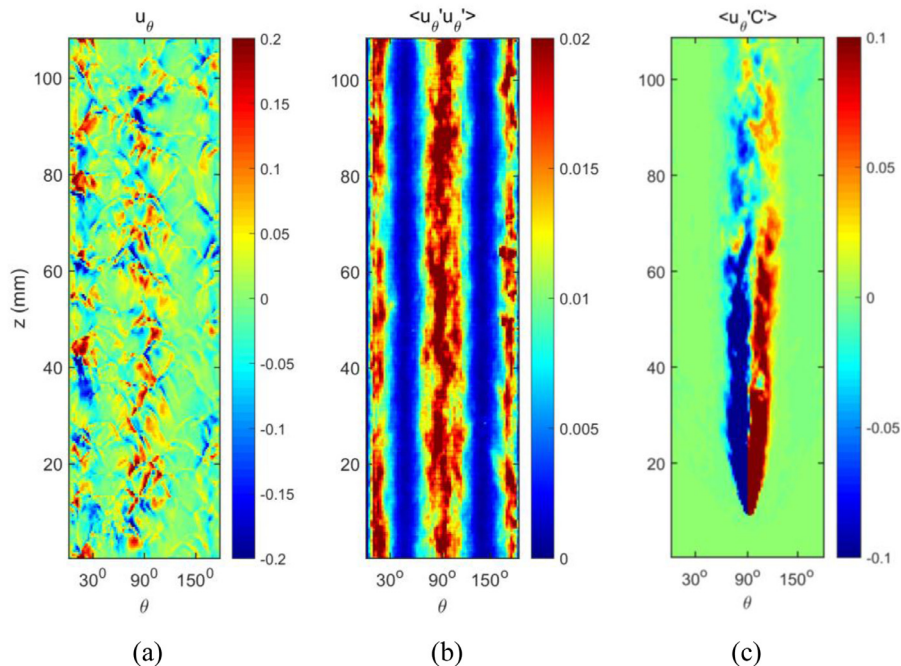


Fig. 24. (a) Instantaneous transverse velocity and (b) variance of transverse velocity and (c) transverse tracer flux in liquid film (LES).

Large fluctuations of the transverse velocity of the liquid film (averaged over film thickness) were recorded at the subchannel gap location (see Fig. 24(b)). These fluctuations are responsible for the enhanced transport of the tracer in the azimuthal direction across the rod. This finding is consistent with the high turbulent activity observed in the subchannel gap region. The cross-movement of the gas-phase, through the action of shear stresses, causes fluctuations in the azimuthal mass transport of the film locally. The absence of these large eddies in URANS results in an under-prediction of the passive scalar mixing. Hence, a thermal-hydraulic analysis of annular flow in a rod bundle geometry should take into account turbulent mixing effects across subchannels.

4. Conclusions

Unsteady two-phase CFD simulations of the annular flow regime were performed and validated against both phase-distribution and tracer mixing experiments in a geometry relevant to industrial applications. The numerical analysis was made possible by simulating flow in a periodic domain by the development of a novel method capable of recycling the outflow boundary conditions. The LES prediction for the time-averaged film thickness distribution was in good agreement with the experiments although some large wavy structures were found to be under-predicted. A non-linear eddy viscosity turbulence model, namely the SST $k-\omega$, QCR model, outperformed the linear realizable $k-\varepsilon$ model by capturing the secondary flow effects in the subchannel geometry. The liquid film flow structure was found to differ from a single-phase turbulent boundary layer due to the presence of the gas-liquid interface. A profile for time-averaged liquid film velocity containing an inflection point is proposed in this work to take into account the two shear layers influencing the flow. Improved prediction of tracer mixing in the liquid film was obtained with LES highlighting the importance of modeling the large eddies in the subchannel gap. Increased turbulent activity in the gap connecting two channels, as captured by the LES, has important implications with regards to the heat transfer characteristics under annular flow regime conditions in rod bundle geometry. The scope of this work clearly shows the capability of current CFD models in predicting the void fraction distribution in BWR subchannels and obtaining liquid film flow properties while other areas have been identified for further research.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.ijmultiphaseflow.2019.05.003](https://doi.org/10.1016/j.ijmultiphaseflow.2019.05.003).

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